

Haoqi Yuan

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Education

Peking University - School of Computer Science Ph.D. Candidate *2021.9 – Present*

- Supervisor: Professor Zongqing Lu. Research Area: Reinforcement Learning, Embodied AI
- Twice awarded the Peking University Presidential Scholarship.

Peking University - Turing Class Bachelor's Degree *2017.9 – 2021.7*

- Major: Computer Science and Technology, GPA 3.79/4, rank 14th/232
- Received National Scholarship and Peking University Merit Student Pacesetter.

Internships

BeingBeyond Research Intern *2025 – Present*

- Study scalable learning methods for dexterous manipulation and humanoid robots.

Beijing Academy of Artificial Intelligence (BAAI) Research Intern *2023 – 2025*

- Study efficient open-world agents equipped with RL and Foundation Models.
- Study RL for generalizable dexterous manipulation.

Hyperplane Lab (PKU) Research Intern *2019 – 2021*

- Study generative models and learning physical interactions, advised by Prof. Hao Dong.

Recent Works

DemoGrasp: Universal Dexterous Grasping from a Single Demonstration

Haoqi Yuan, Ziyue Huang, Ye Wang, Chuan Mao, Chaoyi Xu, Zongqing Lu.

(arXiv:2509.22149) An efficient RL framework for universal dexterous grasping, achieving high success rate from simulation to the real world for any dexterous hand.

Being-H0: Vision-Language-Action Pretraining from Large-Scale Human Videos

Hao Luo, Yicheng Feng, Wanpeng Zhang, Sipeng Zheng, Ye Wang, **Haoqi Yuan**, Jiazheng Liu, Chaoyi Xu, Qin Jin, Zongqing Lu.

(arXiv:2507.15597) The first VLA foundation model trained on large-scale human videos and hand motions, efficiently transferring to robotic dexterous manipulation tasks.

Being-0: A Humanoid Robotic Agent with Vision-Language Models and Modular Skills

Haoqi Yuan, Yu Bai, Yuhui Fu, Bohan Zhou, Yicheng Feng, Xinrun Xu, Yi Zhan, Börje F. Karlsson, Zongqing Lu.

(arXiv:2503.12533) We propose a hierarchical agent framework for humanoid robots, solving long-horizon, embodied tasks autonomously.

Publications

Cross-Embodiment Dexterous Grasping with Reinforcement Learning.

Haoqi Yuan, Bohan Zhou, Yuhui Fu, Zongqing Lu.

ICLR 2025. (First Author)

We propose a reinforcement learning method for training cross-embodiment dexterous grasping policies.

Pre-Trained Multi-Goal Transformers with Prompt Optimization for Efficient Online Adaptation.

Haoqi Yuan, Yuhui Fu, Feiyang Xie, Zongqing Lu.

NeurIPS 2024. (First Author)

We propose a pre-training and fine-tuning method for reinforcement learning that efficiently explores and adapts to long-horizon tasks.

Pre-Training Goal-Based Models for Sample-Efficient Reinforcement Learning.

Haoqi Yuan, Zhancun Mu, Feiyang Xie, Zongqing Lu.

ICLR 2024 (Oral, top 1.2%). (First Author)

We propose pre-training goal-based models to efficiently learn bi-level policies for solving long-horizon tasks in open-world environments.

Plan4MC: Skill Reinforcement Learning and Planning for Open-World Minecraft Tasks.

Haoqi Yuan, Chi Zhang, Hongcheng Wang, Feiyang Xie, Penglin Cai, Hao Dong, Zongqing Lu.

NeurIPS 2023 Workshop. (First Author)

We develop an open-world agent that integrates LLM-assisted planning and reinforcement learning, solving 40 challenging tasks in Minecraft.

Robust Task Representations for Offline Meta-Reinforcement Learning via Contrastive Learning.

Haoqi Yuan, Zongqing Lu.

ICML 2022. (First Author)

We proposed a task representation method combining contrastive learning and generative models to address distribution shift issues in offline meta-reinforcement learning.

DMotion: Robotic Visuomotor Control with Unsupervised Forward Model Learned from Videos.

Haoqi Yuan, Ruihai Wu, Andrew Zhao, Haipeng Zhang, Zihan Ding, Hao Dong.

IROS 2021. (First Author)

We present a data-efficient approach for model-based control by learning motion representations and world models from unsupervised videos.

Learning Diverse Bimanual Dexterous Manipulation Skills from Human Demonstrations.

Bohan Zhou, **Haoqi Yuan**, Yuhui Fu, Zongqing Lu.

AAAI 2025 (Oral).

Efficient Residual Learning with Mixture-of-Experts for Universal Dexterous Grasping.

Ziye Huang, **Haoqi Yuan**, Yuhui Fu, Zongqing Lu.

ICLR 2025.

RL-GPT: Integrating Reinforcement Learning and Code-as-policy.

Shaoteng Liu, **Haoqi Yuan**, Minda Hu, Yanwei Li, Yukang Chen, Shu Liu, Zongqing Lu, Jiaya Jia.

NeurIPS 2024 (Oral).

Creative Agents: Empowering Agents with Imagination for Creative Tasks.

Chi Zhang, Penglin Cai, Yuhui Fu, **Haoqi Yuan**, Zongqing Lu.

UAI 2025.

Awards and Honors

- Peking University Presidential Scholarship (2022, 2021)
- National Scholarship (2018)

- Peking University Merit Student / Pacesetter (2025, 2018)
- Tun Scholarship (2025)
- Luo Yuehua Scholarship (2024)
- Award for Scientific Research, Peking University (2024, 2022)
- Outstanding Student, National Engineering Research Center of Visual Technology (2022)
- John Hopcroft Scholarship (2020)
- Peking University Turing Class Scholarship (2019)
- Second Class Award in Chinese Physics Olympiad (Finals) (2016)

Services

Conference Reviewer

- ICML'22,24,25; NeurIPS'22,23,24,25; ICLR'24,25,26; AAAI'23,24,25,26; CVPR'24,26; CORL'25; IROS'25.

Teaching Assistant

- Deep Reinforcement Learning (Prof. Zongqing Lu, 2023 Spring)
- Computational Thinking in Social Science (Prof. Xiaoming Li, 2020 Autumn)
- Deep Generative Models (Prof. Hao Dong, 2020 Spring)